

5.1.1 BINARY AND IPV4 ADDRESSES

Understanding Binary IP Addressing

1 WHY BINARY?



- Computers, hosts, servers, and network devices use binary addressing to identify each other.
- Binary is a numbering system that uses only two digits: 0 and 1 (called bits).
- IPv4 addresses are written in binary.

BINARY vs DECIMAL

Binary (Base 2)



Uses only 0 and 1

Decimal (Base 10)



Uses digits 0 to 9



2 IPV4 ADDRESS STRUCTURE

An IPv4 address is a 32-bit (4-byte) number written in binary. It is divided into four parts called octets.

IPV4 ADDRESS FORMAT

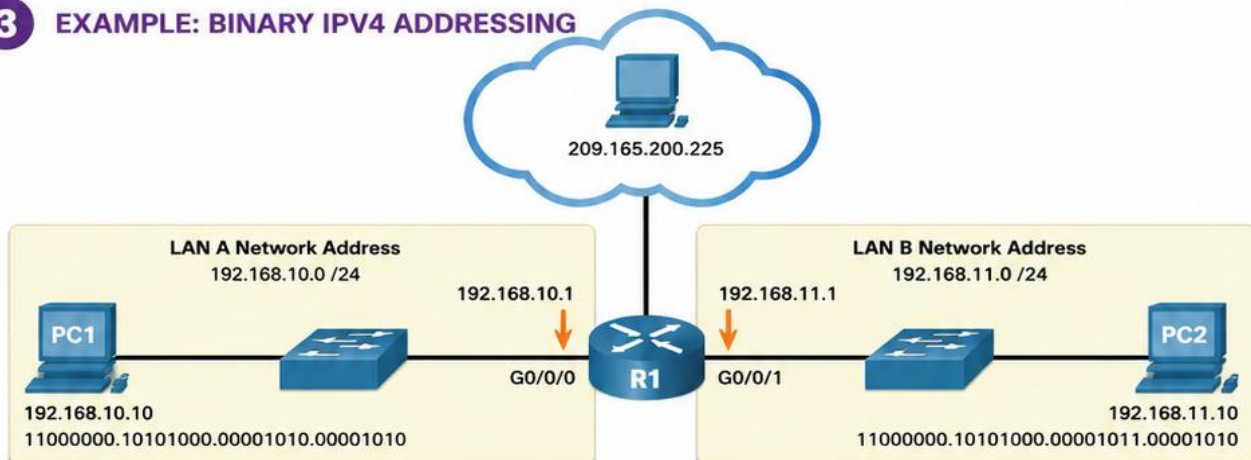


OCTET



- Each octet has 8 bits (1 byte).
- Octets are separated by a dot (.) in dotted decimal notation.
- Valid range of each octet: 0 – 255 (because $2^8 = 256$ values: 0 to 255)

3 EXAMPLE: BINARY IPV4 ADDRESSING



ADDRESS EXAMPLES IN BINARY AND DOTTED DECIMAL

Device / Interface	Address (Binary) → Dotted Decimal
PC1	11000000.10101000.00001010.00001010 → 192.168.10.10
R1 – G0/0/0 (LAN A)	11000000.10101000.00001010.00000001 → 192.168.10.1
R1 – G0/0/1 (LAN B)	11000000.10101000.00001011.00000001 → 192.168.11.1
PC2	11000000.10101000.00001011.00001010 → 192.168.11.10
Cloud (Internet)	11010001.10100101.11001000.11100001 → 209.165.200.225



KEY TAKEAWAYS

- IPv4 address = 32 bits (4 octets).
- Each octet = 8 bits (1 byte).
- Binary uses only 0 and 1.
- Dotted decimal is easier for humans to read (e.g., 192.168.10.10).
- Devices on a network use IP addresses to find and communicate with each other.

4 WHY CONVERT TO DOTTED DECIMAL?



→ 192.168.10.1

- ✓ Binary (32 bits) is difficult for humans to manage.
- ✓ Dotted decimal notation is simple and easy to remember.
- ✓ Used everywhere in networking (configurations, troubleshooting, documentation).



IN SUMMARY: IPv4 addressing is based on binary numbers. Each address has 32 bits (4 octets). We convert binary to dotted decimal (e.g., 192.168.1.1) for human convenience.

5.1.3 BINARY POSITIONAL NOTATION

Understanding the Place Value System in Binary



- A digit's value depends on its position in a number.
- Each position represents a power of the radix (number base).
- Binary uses base 2 (radix = 2). Decimal uses base 10 (radix = 10).

1 DECIMAL POSITIONAL NOTATION (BASE 10)

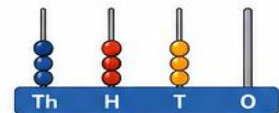
- Each position (place) has a value that is 10 raised to a power.
- From right to left, positions start at 0.

Radix (Base)	10	10	10	10
Position in Number	3	2	1	0
Calculate (10^n)	10^3	10^2	10^1	10^0
Position Value	1000	100	10	1

EXAMPLE:

Decimal number 1234

	Thousands	Hundreds	Tens	Ones
	1000	100	10	1
Decimal Digit (1234)		1	2	4
Calculate	1×1000	2×100	4×10	4×1
Add them up...	1000	+ 200	+ 40	+ 4
Result	1,234			



Th = Thousands (1000)
H = Hundreds (100)
T = Tens (10)
O = Ones (1)

2 BINARY POSITIONAL NOTATION (BASE 2)

- Each position has a value that is 2 raised to a power.
- From right to left, positions start at 0.
- Values grow by powers of 2.

Radix (Base)	2	2	2	2	2	2	2
Position in Number	7	6	5	4	3	2	0
Calculate (2^n)	2^7	2^6	2^5	2^4	2^3	2^2	2^0
Position Value	128	64	32	16	8	4	2

EXAMPLE:

Binary number 11000000

Position Value	128	64	32	16	8	4	2	1
Binary Digit (11000000)	1	1	0	0	0	0	0	0
Calculate	1×128	1×64	0×32	0×16	0×8	0×4	0×2	0×1
Add them up...	128	+ 64	+ 0	+ 0	+ 0	+ 0	+ 0	+ 0
Result	192							



Binary 11000000 equals decimal 192.
(128 + 64 = 192)

3 KEY POINTS

2ⁿ

Position matters

A digit's value depends on its position (place value).



From right to left

Positions start at 0 (least significant bit) and increase leftward.



Powers of the base

Decimal uses powers of 10 (10^0 , 10^1 , 10^2 ...).
Binary uses powers of 2 (2^0 , 2^1 , 2^2 ...).

123

Convert by multiplying

Multiply each digit by its position value, then add all results.

QUICK REFERENCE

Decimal (Base 10) Place Values

1000	(10^3)
100	(10^2)
10	(10^1)
1	(10^0)

Binary (Base 2) Place Values

128	(2^7)
64	(2^6)
32	(2^5)
16	(2^4)
8	(2^3)
4	(2^2)
2	(2^1)
1	(2^0)



TAKEAWAY: Binary positional notation uses base 2, where each position represents a power of 2.

This system allows computers and network devices to represent and process data efficiently.

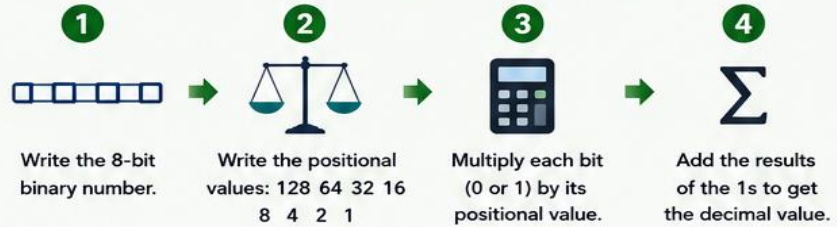
5.1.5 CONVERT BINARY TO DECIMAL



KEY CONCEPT

- An IPv4 address is 32 bits long.
- It is divided into four 8-bit octets.
- Each octet is converted from binary to decimal.
- Use binary positional values and add the values of bits that are 1.

HOW TO CONVERT (FOR ONE OCTET)



Positional Values (for 8-bit octet): 128 64 32 16 8 4 2 1
These values are powers of 2: 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

1 EXAMPLE 1

Convert the first octet of **11000000** to decimal.

★ Reminder

Only the positions with 1 contribute to the total.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (11000000)	1	1	0	0	0	0	0	0
Calculate	1×128	1×64	0×32	0×16	0×8	0×4	0×2	0×1
Add Them Up...	128	+64	+0	+0	+0	+0	+0	+0
Result	192							

✓ So, $11000000_2 = 192_{10}$

2 EXAMPLE 2

Convert the second octet of **10101000** to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (10101000)	1	0	1	0	1	0	0	0
Calculate	1×128	0×64	1×32	0×16	1×8	0×4	0×2	0×1
Add Them Up...	128	+0	+32	+0	+8	+0	+0	+0
Result	168							

✓ So, $10101000_2 = 168_{10}$

3 EXAMPLE 3

Convert the third octet of **00001011** to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (00001011)	0	0	0	0	1	0	1	1
Calculate	0×128	0×64	0×32	0×16	1×8	0×4	1×2	1×1
Add Them Up...	0	+0	+0	+0	+8	+0	+2	+1
Result	11							

✓ So, $00001011_2 = 11_{10}$

4 EXAMPLE 4

Convert the fourth octet of **00001010** to decimal.

Positional Value	128	64	32	16	8	4	2	1
Binary Number (00001010)	0	0	0	0	1	0	1	0
Calculate	0×128	0×64	0×32	0×16	1×8	0×4	1×2	0×1
Add Them Up...	0	+0	+0	+0	+8	+0	+2	+0
Result	10							

✓ So, $00001010_2 = 10_{10}$

COMPLETE IPv4 ADDRESS EXAMPLE



Binary IPv4 Address (32 bits)

11000000 → 192
10101000 → 168
00001011 → 11
00001010 → 10

Decimal IPv4 Address

192.168.11.10

Key Takeaway

Convert each 8-bit octet independently using positional values (powers of 2) and add the values where the bit is 1.



QUICK RECAP

- ✓ IPv4 address = 4 octets $\times$ 8 bits = 32 bits
- ✓ Binary to decimal: Multiply each bit by its positional value and add.
- ✓ Positional values (8-bit): 128, 64, 32, 16, 8, 4, 2, 1
- ✓ Example: $11000000.10101000.00001011.00001010 = 192.168.11.10$

5.1.7–5.1.11 BINARY NUMBER SYSTEM (PART 4)

Decimal ↔ Binary Conversion and IPv4 Addresses

1 5.1.7 DECIMAL TO BINARY CONVERSION



A useful tool is the binary positional value table (powers of 2).

BINARY POSITIONAL VALUE TABLE (8-BIT OCTET)

Position Value	128	64	32	16	8	4	2	1
2 ⁿ (Power of 2)	2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰

How it works (Left to Right):

- For each position, if the decimal number is ≥ the position value, write 1 and subtract that value from the number.
- If the decimal number is < the position value, write 0.
- Continue until all positions are processed.



KEY TAKEAWAYS

- Use the 8-bit positional value table (128 to 1).
- Compare the decimal number with each value from left to right.
- Write 1 if value is used, 0 if not.
- Subtract used value and continue.
- Result is an 8-bit binary number.

EXAMPLE: Convert 192.168.10.11 to Binary

1

First Octet: 192

Pos. Value	128	64	32	16	8	4	2	1
192 (Decimal)	1	1	0	0	0	0	0	0
Subtract	192-128=64							

→ Result: 11000000₂ = 192₁₀

2

Second Octet: 168

Pos. Value	128	64	32	16	8	4	2	1
168 (Decimal)	1	0	1	0	1	0	0	0
Subtract	168-128=40		40-32=8					

→ Result: 10101000₂ = 168₁₀

3

Third Octet: 10

Pos. Value	128	64	32	16	8	4	2	1
10 (Decimal)	0	0	0	0	1	0	1	1
Subtract					10-8=2		2-2=0	

→ Result: 00001010₂ = 10₁₀

4

Fourth Octet: 11

Pos. Value	128	64	32	16	8	4	2	1
11 (Decimal)	0	0	0	0	1	0	1	1
Subtract					10-8=3		3-2=1	1-1=0

→ Result: 00001011₂ = 11₁₀



Final Binary IPv4 Address

11000000.10101000.00001010.00001011

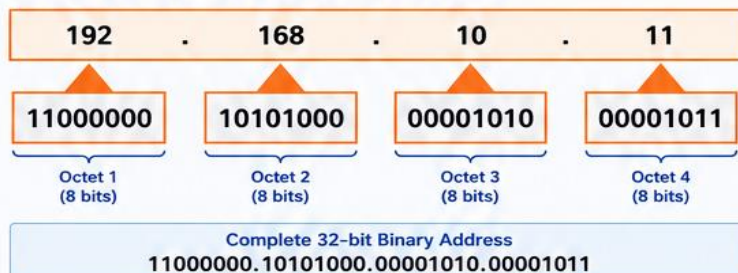
2 5.1.11 IPv4 ADDRESSES

DOTTED DECIMAL ADDRESS

OCTETS

32-BIT ADDRESS

An IPv4 address has 4 octets (8 bits each). Computers store it as a 32-bit binary number.



3 5.1.8 DECIMAL TO BINARY CONVERSION EXAMPLE (SUMMARY)

Octet	Decimal	Binary (8 bits)	Decimal → Binary	Binary → Decimal Check
1st Octet	192	11000000	192 = 128 + 64	128 + 64 = 192
2nd Octet	168	10101000	168 = 128 + 32 + 8	128 + 32 + 8 = 168
3rd Octet	10	00001010	10 = 8 + 2	8 + 2 = 10
4th Octet	11	00001011	11 = 8 + 2 + 1	8 + 2 + 1 = 11
Final IPv4 Address		192.168.10.11	11000000.10101000.00001010.00001011	



REMEMBER

- 1 octet = 8 bits
- 4 octets = 32 bits
- Dotted decimal is for humans
- Binary is used by computers
- Practice makes conversion easy!



QUICK RECAP

- ✓ Decimal → Binary: Use positional values (128 to 1) and subtract.
- ✓ Binary → Decimal: Multiply each bit by its positional value and add.

- ✓ IPv4 Address: 4 octets, each 8 bits = 32 bits total.
- ✓ Example: 192.168.10.11 = 11000000.10101000.00001010.00001011

5.2.1 – 5.2.4 HEXADECIMAL NUMBER SYSTEM (PART 4)

Hexadecimal, IPv6 and Conversions

1 5.2.1 HEXADECIMAL AND IPv6 ADDRESSES



- Hexadecimal is a base-16 number system.
- It uses digits 0–9 and letters A–F.
- It is easier to express a value as a single hex digit than four binary bits.
- Used in networking for IPv6 addresses and Ethernet MAC addresses.

DECIMAL, BINARY AND HEXADECIMAL EQUIVALENT

Decimal (Base 10)	Binary (Base 2)	Hexadecimal (Base 16)
0	0000	0
1	0001	1
2	0010	2
...	...	...
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F



- IPv6 address length = 128 bits.
- Every 4 bits (a single hex digit) has 16 possible values (0–F).
- Total IPv6 values = 16^{32} (because 128 bits = 32 hex digits).
- IPv6 is not case-sensitive (a–f or A–F both valid).

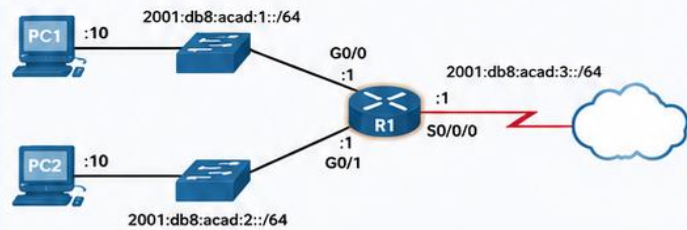
2 5.2.1 IPv6 ADDRESS FORMAT

- Preferred format: x:x:x:x:x:x:x:x
- Each "x" is one hex digit (4 bits).
- When referring to 8 bits (1 octet) in IPv4, we use the term octet.
- In IPv6, a *hextet* is used to refer to 16 bits (4 hex digits).



8 hextets = 32 hex digits = 128 bits

SAMPLE IPv6 TOPOLOGY



3 5.2.3 DECIMAL TO HEXADECIMAL CONVERSIONS

STEPS



- Convert the decimal number to 8-bit binary.
- Divide the 8-bit binary into groups of four bits from right to left.
- Convert each 4-bit group to its hexadecimal digit.

EXAMPLE: Convert 168_{10} to Hexadecimal

- 168 in binary (8-bit)
10101000
- Group into 4-bit sets
1010 1000
- Convert each group to hex digit
A 8

Answer:
 $168_{10} = A8_{16}$
A8

BINARY ↔ HEX QUICK MAP

0000 = 0	1000 = 8
0001 = 1	1001 = 9
0010 = 2	1010 = A
0011 = 3	1011 = B
0100 = 4	1100 = C
0101 = 5	1101 = D
0110 = 6	1110 = E
0111 = 7	1111 = F

4 5.2.4 HEXADECIMAL TO DECIMAL CONVERSIONS

STEPS



- Convert each hex digit to 4-bit binary.
- Group the binary into 8-bit sets from right to left.
- Convert each 8-bit set to decimal.

EXAMPLE: Convert $D2_{16}$ to Decimal

- Hex to 4-bit binary
D 2
1101 0010
- Group into 8-bit set
11010010
- Convert to decimal
 $11010010_2 = 210_{10}$

Answer:
 $D2_{16} = 210_{10}$
210



NOTES

- 1 hex digit = 4 bits
- 1 hextet (4 hex digits) = 16 bits
- IPv6 total = 8 hextets = 128 bits
- Hex makes addresses shorter and easier to read than binary.

SUMMARY OF KEY POINTS

NUMBER SYSTEMS

2 10 16

- Binary (Base 2): 0, 1
- Decimal (Base 10): 0–9
- Hexadecimal (Base 16): 0–9, A–F

WHY HEXADECIMAL?

$1010_2 \rightarrow A_{16}$

- 4 bits binary = 1 hex digit
- Shorter representation
- Common in IPv6 and MAC addresses

IPv6 FORMAT

x:x:x:x:x:x:x:x

- Each x = 4 hex digits (16 bits)
- 8 hextets = 32 hex digits
- Total = 128 bits

CONVERSION SUMMARY



- Decimal → Hex: Decimal → 8-bit binary → group 4 bits → to hex
- Hex → Decimal: Hex → 4-bit binary → group 8 bits → to decimal



QUICK FACTS

- Hextet = 16 bits
- Example: $D2_{16} = 210_{10}$
- IPv6 addresses are not case-sensitive